Operationally Cohesive QUBO Formulations for Real-Time Air Traffic Deconfliction Using Probabilistic Computing

Boris Baros

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1 Initial Plan (Notes)

1.1 Key Works

Quantum Annealing Applied to De-Conflicting Optimal Trajectories [6]

- Provides a QUBO formulation of the problem on a fleet-level. Only ground-holding is permitted as an action.
- Formulates the conflict-based graph to sparsify dependence graph.
- Notes significant increase in difficulty as more agents enter the space.
- Minimises total delay.

Hybrid Quantum-Classical Multi-Agent Pathfinding [3]

- Uses a column generation approach to slowly increase the path space, to increase efficiency.
- Introduces a theoretically sound stopping criterion for when it is optimal.
- Provides extensive results on very dense, 2D scenarios.
- Action spaces are vastly simplified (up, down, left and right moves).

1.2 Motivations

There are some clear ways in which we can improve on the aforementioned works, generally thanks to proper consideration of operational constraints/realities:

- Thanks to the always-on nature of ATC, deconfliction doesn't really happen on a fleet level, all at once. Rather, it is a more first-come, first-served situation. This gives us a clear fairness criterion, but also simplifies matters: we should rather consider how to optimally "slot in" a flight, rather than perturbing every flight for deconfliction (which is operationally unrealistic). This will also allow us to simplify complexities arising from a high number of agents, as we will aim to keep most flights fixed.
- A total-delay minimisation isn't necessarily fair, as some flights may be more significantly affected than others in order to achieve a globally minimised delay. To this end, we explore a minimax formulation of the problem, where we wish to minimise the maximal delay across all perturbed flights. This ensures we aim to provide a bound on delays, which is operationally desirable to airlines, giving them confidence that their schedules will not be compromised disproportionately.

- Taking an adaptive approach to the problem will allow us to complexify the action spaces. This is ideal, as an emulation of ATCO decisions is not simple, and we will need to think carefully to mirror them with any automated strategy. To that end, in our framework we will consider alternative routes and multiple levels (with alternative routing staying around the proposed trajectory). Our hope is that this action space will be rich enough to consistently meet deconfliction standards. We do not change velocities as small changes are allowed in practice, and timescales could be too short for this to make enough difference. Further, we do not explore ground-holding like in [6], as we would like to isolate the en-route phase without offloading difficulties onto the airport-specific optimisation problems of managing queues for departures.

- The nature of the individual cost function of a flight is unbeknownst to the controller. Noting further the fact that fuel consumption evolves highly nonlinearly with changes in the path, there is no guarantee that a small perturbation in a path will lead to a small change in the resulting fuel cost - therefore, slight perturbations on the sphere of the optimal path are probably not desirable manoeuvres. On the other hand, we cannot compute another optimal route. It is not unreasonable to expect a pilot to wish to commit to another path that they provide themselves. Therefore, we consider rerouting in the sense of either staying to the current path or deviating onto another requested one from potentially multiple options.

- Current approaches don't provide much nuance to time taken to traverse a route, in the sense of incorporating weather constraints. This is a complexity that would render a multi-agent scenario even more computationally challenging. Our adapted variation, where flight-specific optimal routes are already provided, would only need to store weather data along the routes being perturbed, which is much more efficient. Further, this simplification would allow us to incorporate uncertainties, to then ensure robustness to forecast inaccuracies/ uncertainties.

- The column generation approach from [3] could be a good way to investigate increasing the planes potentially perturbed.

2 Formulating the Optimisation Problem

We consider the following situation. Suppose that there are currently N flights, which may or may not be en-route, whose trajectories we assume to be deconflicted. Their trajectories (considered to be optimal for their respective airline cost indices) are given by $\xi^{(i)} := (\phi_t^{(i)}, \lambda_t^{(i)}, h_t^{(i)})_{t=\tau_0^{(i)}}^{\tau_1^{(i)}} \subset [-\frac{\pi}{2}, \frac{\pi}{2}] \times [-\pi, \pi] \times H_{\text{legal}}^{(i)}$, which a time-discretised path (in longitude-latitude-altitude coordinates along the waypoints) from position $\xi_{\tau_0^{(i)}}^{(i)}$ at departure time $t = \tau_0^{(i)}$, all the way through to the arrival point $\xi_{\tau_1^{(i)}}^{(i)}$, scheduled at time $\tau_1^{(i)}$. As described in [5], the set of potential altitudes is first decided via a vertical profile optimisation, followed by projection onto the permitted flight levels, given by the finite, discrete set $H_{\text{legal}}^{(i)}$ (the maximal flight level could be airplane-specific). The flight plan will also contain a chosen speed of flight $m^{(i)}$ (or multiple at different waypoints), whose units here we assume to be the Mach number, so that the True Airspeed (TAS) is the Mach number multiplied by the local speed of sound.

For the sake of our optimisation procedure, we convert the given trajectories into a set of artificial-time indexed trajectories, denoted by $(\xi_\tau^{(i)})_{\tau=0}^{T^{(i)}}$, where $T^{(i)} + 1$ denotes the number of waypoints of the given trajectory (when including the endpoints). We begin by outlining the decision sets below, which includes careful consideration of constructing a trajectory-local spatial grid of potential deviations, rather than laying everything out on a global spatial grid.

2.1 Defining the Action Space

In order to attach clear connection to the concept of minimal deviation local to the given (optimal) trajectory, we introduce a trajectory-local grid construction for interpreting potential redirection manoeuvres. We will begin by considering one representative trajectory, given by $\xi := (\phi_\tau, \lambda_\tau, h_\tau)_{\tau=0}^{T}$, where $T+1$ is the number of waypoints. Except for the departure and arrival waypoints, at each waypoint we define a set of potential lateral deviation waypoints, parametrised by the separation distance d and number K that we wish to allow. Indeed, each waypoint $(\phi_\tau, \lambda_\tau)$ can be written as a Rodrigues rotation from the previous waypoint via

$$\hat{x}(\phi_\tau, \lambda_\tau) = \cos(\delta_\tau)\hat{x}(\phi_{\tau-1}, \lambda_{\tau-1}) + \sin(\delta_\tau)\hat{d}_\tau,$$

where:

- $\hat{x} : [-\frac{\pi}{2}, \frac{\pi}{2}] \times [-\pi, \pi] \to \mathbb{S}^2$ denotes the mapping from latitude-longitude pair to a 3D vector on the unit sphere, given by

$$\hat{x}(\phi, \lambda) = \begin{pmatrix} \cos\phi\cos\lambda \\ \cos\phi\sin\lambda \\ \sin\phi \end{pmatrix};$$

- δ_τ denotes the angular distance between the points at waypoints $\tau - 1$ and τ, computed by

$$\delta_\tau = \arccos\big(\hat{x}(\phi_{\tau-1}, \lambda_{\tau-1}) \cdot \hat{x}(\phi_\tau, \lambda_\tau)\big);$$

- $\hat{d}_\tau$ denotes the unit vector tangent to the unit sphere at waypoint $\tau - 1$, pointing towards the waypoint at τ, defined by rearranging the above,

$$\hat{d}_\tau = \frac{\hat{x}(\phi_\tau, \lambda_\tau)}{\sin(\delta_\tau)} - \frac{\hat{x}(\phi_{\tau-1}, \lambda_{\tau-1})}{\tan(\delta_\tau)}.$$

Since, at waypoint τ the direction of rotation jumps from $\hat{d}_\tau$ to $\hat{d}_{\tau+1}$, we take the bisecting direction given by

$$\hat{t}_\tau := \frac{1}{2}\big(\hat{d}_\tau + \hat{d}_{\tau+1}\big),$$

and compute the tangent vector that is perpendicular, denoted

$$n_\tau := \hat{t}_\tau \times \hat{x}(\phi_\tau, \lambda_\tau).$$

Using this, at waypoint τ we can define the potential lateral deviation waypoints to be

$$W(\tau, K, d, \xi) := \Big\{ \cos\Big(\frac{kd}{R}\Big)\hat{x}(\phi_\tau, \lambda_\tau) + \sin\Big(\frac{kd}{R}\Big)n_\tau \Big\}_{k=-K}^{K},$$

where R is an approximate radius for the Earth.

In this sense, we have produced a skeletal deviation structure that is moulded to the optimal trajectory, inspired by the wind-aware OTS structures conceptualised in [2]. We traverse this structure by specifying a sequence of deviation variables, denoted $(\theta_\tau)_{\tau=1}^{T-1} \subset \{-K, -K+1, \ldots, K-1, K\}$, which state which element of the deviation structure $W(\tau, K, d, \xi)$ to occupy at waypoint τ, with the convention that the start and end deviations are fixed as nominal, $\theta_0 = \theta_T = 0$. Then, if we specify $\theta_\tau = 0$ for all τ, then we maintain the original proposed trajectory, otherwise the deviated trajectory is recovered by

$$\Big(W(\tau, K, d, \xi)_{\theta_\tau}\Big)_{\tau=0}^{T}.$$

In a similar fashion to the above, there exists a vertical flight profile that the controller would like to adhere to, given by $(h_\tau)_{\tau=1}^{T-1}$. As reflected by the success metrics of ATCOs, remaining within a 2-flight-level-band of this profile at all times is considered sufficient adherence to the requested vertical profile, but we will generalise here to a J-flight-level-band for generality. Hence, we have the vertical trajectory-adhering structure given by

$$H(\tau, H_{\text{legal}}, \xi) := \{h\}_{h=(h_\tau - J)\vee\inf\{H_{\text{legal}}\}}^{(h_\tau+J)\wedge\sup\{H_{\text{legal}}\}}.$$

At every waypoint τ, we issue a flight level deviation $\chi_\tau \in \{-J, -J+1, \ldots, J-1, J\}$ reflecting the level relative to h_τ. To ensure that the resulting flight level $\chi_\tau + h_\tau$ remains within the permissible level set $H(\tau, H_{\text{legal}}, \xi)$, we add constraints to remain within H_{legal}. We will eventually discuss how to constrain the complexity of the full manoeuvre process $(\theta_\tau, \chi_\tau)_{\tau=1}^{T-1}$, but for now we proceed to discuss computation of costs for the edges constituting this structure.

2.2 Estimating Edge Costs

For every perturbable flight i, we enumerate a trajectory-local graph, whose edges are characterised by the deviation instructions discussed above. Concretely, for perturbable flight i, we define a spatiotemporal undirected graph $G^{(i)} := (V_i, E_i)$, where $V_i := \{0, 1, \ldots, T^{(i)}\} \times V, E_i := \{0, 1, \ldots, T^{(i)}\} \times E$ denote the time-enriched versions of the static vertex and edge options, V, E. Vertices $v \in V_i$ of the full graph are of the form

$$v = (\tau, \theta, \chi),$$

and edges have the feedforward form

$$e = (v, v') = \Big((\tau - 1, \theta, \chi), (\tau, \theta', \chi')\Big) \in V_i \times V_i,$$

where $(\theta, \chi) \in \{-K, -K+1, \ldots, K-1, K\} \times \{-J, -J+1, \ldots, J-1, J\}$ indicate lateral deviation θ and flight level deviation χ at waypoint $\tau - 1$.

Returning to our action space above, the aim of the controller is to leverage these actions at each waypoint to optimally navigate from the starting vertex $v_0^{(i)} = (0, 0, 0)$ to $v_{T^{(i)}} = (T^{(i)}, 0, 0)$, given some objective and constraints that we specify later, via some intermediate path $((\tau, \theta_\tau^{(i)}, \chi_\tau^{(i)}))_{\tau=1}^{T^{(i)}-1}$. Though time-optimality will play a role, it will not form the primary objective of our framework - it is in this sense that our formulation is not a standard shortest-path problem. Nevertheless, we still associate to each edge a cost corresponding to the time to traverse the space. Since the graph is feedforward in waypoint-time, we denote these costs as $c_i(\tau, \theta_{\tau-1}^{(i)}, \chi_{\tau-1}^{(i)}, \theta_\tau^{(i)}, \chi_\tau^{(i)})$, indicating the time it takes flight i to traverse from lateral-vertical deviation pair $(\theta_{\tau-1}^{(i)}, \chi_{\tau-1}^{(i)})$ at waypoint $\tau - 1$ to $(\theta_\tau^{(i)}, \chi_\tau^{(i)})$ at waypoint τ.

We precompute these and then fix them as edge costs in our graphical representation. Below we outline how to compute these estimates, using a forecast given by Z.

2.2.1 Constant-Level Edge Costs

To estimate the timepoints of a trajectory we need to discretise along it, which only gives estimated times for each spatial point in the discretisation. Arguably, this is one of the main obstacles in the literature: incorporating the highly nonlinear wind field forecasts. This requires storing much weather forecast information and fine discretisation for accurate estimation. This precomputation step renders any full-grid search space (such as that of [3]) infeasible for pathfinding.

The trajectories between waypoints follow a great-circle path, so that we may write any point $\mathbf{x}(\alpha)$ on the path from $(\phi_\tau, \lambda_\tau)$ to $(\phi_{\tau+1}, \lambda_{\tau+1})$ using the Rodrigues rotation formula,

$$\mathbf{x}(\alpha) = \cos(\alpha)\hat{x}(\phi_\tau, \lambda_\tau) + \sin(\alpha)\hat{d}_\tau, \quad \alpha \in [0, \delta_\tau],$$

where the $\hat{x}, \delta_\tau, \hat{d}_\tau$ objects are as defined in Section 2.1. Suppose then that we have access to a forecast, which we call Z, and that we split the great circle between each waypoint pair into M subintervals. This gives a refined set of waypoints, denoted $(\phi_l, \lambda_l)_{l=0}^{TM}$, where

$$\hat{x}(\phi_l, \lambda_l) = \cos\left(\delta_{\lfloor l/M \rfloor}\left(\frac{l}{M} - \left\lfloor \frac{l}{M} \right\rfloor\right)\right)\hat{x}(\phi_{\lfloor l/M \rfloor}, \lambda_{\lfloor l/M \rfloor}) + \sin\left(\delta_{\lfloor l/M \rfloor}\left(\frac{l}{M} - \left\lfloor \frac{l}{M} \right\rfloor\right)\right)\hat{d}_{\lfloor l/M \rfloor}.$$

Denote now the relevant $\mathbb{S}^2$ waypoints by $(\hat{x}_l)_{l=0}^{TM}$. We compute the angular speed from $\hat{x}_l$ to $\hat{x}_{l+1}$ via

$$\omega_l = \sqrt{(v_l a_\infty)^2 - \|Z_{\text{cross}}(\hat{x}_l)\|^2} + Z_{\text{tail}}(\hat{x}_l) \cdot \hat{x}_l,$$

where the parallel and perpendicular components from Z are computed using the meridional and zonal components (u, v respectively),

$$Z_{\text{tail}}(\hat{x}_l) := \left(\left[v(x)e_N(x) + u(x)e_E(x)\right]\Big|_{x=\hat{x}^{-1}(\hat{x}_l)} \cdot \frac{\mathrm{d}}{\mathrm{d}\alpha}\hat{x}_l\right)\frac{\mathrm{d}}{\mathrm{d}\alpha}\hat{x}_l,$$
$$Z_{\text{cross}}(\hat{x}_l) := \left[v(x)e_N(x) + u(x)e_E(x)\right]\Big|_{x=\hat{x}^{-1}(\hat{x}_l)} - Z_{\text{tail}}(\hat{x}_l),$$

where $\hat{x}^{-1}$ denotes the transformation back to latitude-longitude coordinates, and e_N, e_E denote the local tangential North-East basis in spherical coordinates, computed using

$$e_N(\phi, \lambda) = \begin{pmatrix} -\sin\phi\cos\lambda \\ -\sin\phi\sin\lambda \\ \cos\phi \end{pmatrix}, \quad e_E(\phi, \lambda) = \begin{pmatrix} -\sin\lambda \\ \cos\lambda \\ 0 \end{pmatrix}.$$

With all this information, the time taken to traverse from $\hat{x}_l$ to $\hat{x}_{l+1}$ can be computed as

$$c_{l+1} := \frac{\delta_{\lfloor l/M \rfloor}}{M\omega_l}.$$

We have assumed in the above that M is large enough that speed on the subinterval remains constant. Using this notation, the total duration of the flight is given by

$$t := \sum_{l=1}^{TM} c_l,$$

with the genuine time estimates to clear subinterval N given by

$$\tau_0 + s_N := \tau_0 + \sum_{l=1}^{N} c_l,$$

so that the estimated arrival time is given by $\tau_0 + s_{TM} = \tau_0 + t$.

2.2.2 Level-Transitioning Edge Costs

Write estimation procedure out! BADA model stuff.

2.3 Primary Objective

Informally speaking, our approach should mirror the decision-making of controllers. We posit the following hierarchical structure, assuming $N-1$ current deconflicted flights and attempting to deconflict an incoming N-th flight:

1. Given the current set of perturbable flights, we find the simplest deviation instruction set that will ensure both deconfliction and delays within the allowed interval.

2. Add another flight to the set of perturbable flights, and reattempt the above optimisation.

3. If the total deviation instruction complexity increases, then we terminate at the results of (1). If not, then repeat the process, until termination.

If we can model deviation complexity in a clear way, then the above will lead to sensible compromises between perturbing more flights and further complicating manoeuvres, all while satisfying deconfliction and delay constraints.

Remark 1. *We note that a multi-objective Lagrangian formulation could be used to reflect a balance between complexity arising from deviation instructions, and that of perturbing multiple flights. In fact, the coefficients would have a clear interpretation owing to the integer-valued objectives in question. However, this would not reflect the hard hierarchical decision-making of controllers - that, and the fact that our hierarchical formulation restricts the deviation space to small numbers of flights, rather than considering all flights at once.*

We will begin by formulating the problem for a fixed set of perturbable flights, for which we will need to carefully define confliction regions. Following this, we will present an operationally faithful iterative procedure, which slowly augments the set of perturbable flights until all constraints are met, and total instruction complexity cannot be decreased further.

We suppose that the total set of perturbable flights is given by some $\mathcal{P}\cup\{N\} := \mathcal{P}_{\cup N}$, where $\mathcal{P} \subseteq [N-1]$ denotes the set of perturbable non-N flights (we will discuss how to define this in due course). This endows the controller with decision variables $\{(\theta_\tau^{(i)}, \chi_\tau^{(i)})_{\tau=1}^{T^{(i)}-1}\}_{i\in\mathcal{P}_{\cup N}}$, which correspond to the controller wishing to issue deviation instructions to ensure deconfliction within acceptable arrival time deviations. The controller also wishes to issue the simplest instructions set possible, providing a primary objective of the form

$$\sum_{i\in\mathcal{P}_{\cup N}}\sum_{\tau=1}^{T^{(i)}}\Big(2|\theta_\tau^{(i)}-\theta_{\tau-1}^{(i)}|^2+|\chi_\tau^{(i)}-\chi_{\tau-1}^{(i)}|^2\Big),$$

where we recall that $\theta_0^{(i)}, \chi_0^{(i)}, \theta_{T^{(i)}}^{(i)}, \chi_{T^{(i)}}^{(i)} := 0$ for the fixed endpoints. Before discussing constraints and confliction definitions, we make some remarks around our instruction complexity interpretation.

Remark 2.

- *The lateral and vertical deviation penalties are in terms of differences, rather than the instantaneous choices - this is deliberate. The construction above can be understood as a deviation complexity penalty when one notes the outcome of a significantly oscillatory lateral deviation set. Intuitively, although the oscillatory path might remain locally tight to the preferred trajectory $\xi^{(i)}$, it would represent potentially multiple deviations - we would like to limit the number of deviations in the lateral manoeuvres. In terms of remaining locally tight to the trajectories, this constraint is implicitly imposed by constraining the delay to be close to the schedule - prolonged deviations from proposed trajectories will not be feasible under such a constraint, whereas local ones will.*

- *We have made the seemingly ad-hoc choice of placing a numerical coefficient in front of the lateral deviation penalties. This explicitly encodes the preference to deviate vertically rather than laterally - a standard convention in the decision-making process of the air traffic controllers.*

2.4 Formulating a Potential Conflict Set

In addition to precomputing the edge costs $\{c_i(\tau,\theta_1,\theta_2,h_1,h_2)\}_{\tau,i,\theta_1,\theta_2,h_1,h_2}$ for $i\in\mathcal{P}_{\cup N}$, we must somehow precompute a set of potential conflicts on which to formulate conflict constraints. This

will certainly be more computationally efficient than formulating conflict avoidance constraints between every point on every potential trajectory and every other point on every other trajectory. In [6] it was very simple to do this in a spatiotemporal fashion, as the action set involved explicit delays on a bounded discrete set. In our setting however, where the c_i depend on wind, it is not so clear to biject our set of actions to such a set of delays. Nonetheless, we can define the set of potential conflicts via spatial information, which is fixed beforehand.

We begin by reformulating the trajectory process of flights to that of sequentially occupying subintervals along a chosen trajectory, which are occupied for intervals of time given by the respective edge costs. This not only simplifies proceedings, but also increases robustness. On the other hand, the subintervals given by waypoint-to-waypoint navigation could be potentially quite large, so we split every waypoint gap into δ subintervals, so that a trajectory $\xi^{(i)}$ gets split into $T^{(i)}\delta$ subintervals. We proceed without the additional δ notation, as it only complicates our framework - it is merely a parameter that allows us to control the size of the subintervals being occupied more finely than waypoint-to-waypoint navigation.

Suppose that we have a set of perturbable flights, denoted above by $\mathcal{P}_{\cup N}$. By the very nature of our framework, flights in $\mathcal{P}^c_{\cup N}$ are deconflicted with one another - the only potential conflicts exist between flights in $\mathcal{P}_{\cup N}$, or between flights in $\mathcal{P}_{\cup N}$ with those in $\mathcal{P}^c_{\cup N}$. Hence for all flights in $\mathcal{P}_{\cup N}$, we define some overall spatial subinterval conflict set as

$$C_{\mathcal{P}_{\cup N}} := \bigcup_{i \in \mathcal{P}_{\cup N}} C^{(i)}_{\mathcal{P}_{\cup N}},$$

where each $C^{(i)}_{\mathcal{P}_{\cup N}}$ denotes the pairs of edges, with one edge in deviation structure of flight i, that are spatially conflicting, with the subscript $\mathcal{P}_{\cup N}$ denoting the set of flights for which we compare to all subintervals of the deviation skeleton (those in the set of perturbable flights), and those for which we only consider the given, unperturbable, trajectory. In more detail, we have the characterisation

$$\begin{aligned} C^{(i)}_{\mathcal{P}_{\cup N}} := &\{(u,v) \in E_i \times E_j : j \in \mathcal{P}_{\cup N} \setminus \{i\}, \underline{\mathrm{d}}(u,v) \leq \alpha\} \\ &\cup \{(u,v) \in E_i \times E^{\xi}_j : j \in \mathcal{P}^c_{\cup N}, \underline{\mathrm{d}}(u,v) \leq \alpha\}, \end{aligned} \tag{1}$$

where E_i denotes the enriched edge-set of the deviation skeleton around trajectory $\xi^{(i)}$, α denotes the separation distance standards, $\underline{\mathrm{d}}$ denotes the minimum distance between two edges, and E^{ξ}_j denotes the edge-set for flight j where there is no perturbation, so that vertices are of the form $(\tau, 0, 0)$ for all τ. Given that the edges u, v are great circle arcs, we may explicitly compute $\underline{\mathrm{d}}$. Segments u and v will either intersect, or will achieve their minimal distance from one another from an endpoint of one to the other edge. For the first case, we compute

$$\begin{aligned} \delta_{u,v} := \min\Big\{ &|\delta(u_0, p_{u,v}) + \delta(p_{u,v}, u_1) - \delta(u_0, u_1)| + |\delta(v_0, p_{u,v}) + \delta(p_{u,v}, v_1) - \delta(v_0, v_1)|, \\ &|\delta(u_0, -p_{u,v}) + \delta(-p_{u,v}, u_1) - \delta(u_0, u_1)| + |\delta(v_0, -p_{u,v}) + \delta(-p_{u,v}, v_1) - \delta(v_0, v_1)| \Big\}, \end{aligned}$$

where subscripts 0 and 1 denote the endpoints of edges (ignoring flight levels) converted into Cartesian coordinates on the unit sphere, $\delta(a, b) := \arccos(a \cdot b)$ denotes the angular distance between a and b, and $p_{u,v}, -p_{u,v}$ denote the pair of antipodal points at which the full great circle extensions of u, v intersect, given by

$$p_{u,v} = \frac{n_u \times n_v}{\|n_u \times n_v\|},$$

where

$$n_u := \frac{\hat{x}(u_0) \times \hat{x}(u_1)}{\|\hat{x}(u_0) \times \hat{x}(u_1)\|}$$

denotes the normal to the plane through the origin in which edge u exists. If $\delta_{u,v} = 0$, then it is either the case that $p_{u,v}$ lies on both u, v, or that $-p_{u,v}$ occupies both.

If this is not the case, then the minimal distance is from one of the endpoints to the adjacent arc. Consider some general point $x \in \mathbb{S}^2$, whose minimal distance from u we compute. The unsigned distance between x and the full great circle containing u is given by $R\arcsin(|x \cdot n_u|)$. We then check whether the closest point on the full great circle lies on u by projecting x onto the plane arising from normal n_u, and then projecting to the unit sphere, giving

$$q_{u,x} := \frac{x - (x \cdot n_u)n_u}{\|x - (x \cdot n_u)n_u\|}.$$

As with the condition above for the intersecting case, we have $q_{u,x} \in u$ if and only if

$$|\delta(u_0, q_{u,x}) + \delta(q_{u,x}, u_1) - \delta(u_0, u_1)| = 0.$$

If this is not the case, then the closest point on u is one of the endpoints. Therefore, the shortest distance from x to u is given by

$$\underline{\mathrm{d}}(x, u) := \begin{cases} R\arcsin(|x \cdot n_u|) & \text{if } q_{u,x} \in u, \\ R\min\{\delta(x, u_0), \delta(x, u_1)\} & \text{otherwise.} \end{cases}$$

To summarise, given two edges u, v, we have

$$\underline{\mathrm{d}}(u, v) = \begin{cases} 0 & \text{if } \delta_{u,v} = 0, \\ \min\{\underline{\mathrm{d}}(u_0, v), \underline{\mathrm{d}}(u_1, v), \underline{\mathrm{d}}(u, v_0), \underline{\mathrm{d}}(u, v_1)\} & \text{otherwise,} \end{cases}$$

with which we may compute the spatially conflicting set $C_{\mathcal{P}_{\cup N}}$.

For each flight's trajectory skeleton E_i (or E_j^{ξ} if $j \in \mathcal{P}_{\cup N}^c$), and chosen deviation instructions $\{(\theta_\tau^{(i)}, \chi_\tau^{(i)})\}_{\tau=1}^{T^{(i)}-1}$ (or $\{(0,0)\}_{\tau=1}^{T^{(j)}-1}$ if $j \in \mathcal{P}_{\cup N}^c$), we may deduce an edge occupation process

$$U^{(i)} := (u_\tau^{(i)})_{\tau=1}^{T^{(i)}} := \left(\left((\tau - 1, \theta_{\tau-1}^{(i)}, \chi_{\tau-1}^{(i)}), (\tau, \theta_\tau^{(i)}, \chi_\tau^{(i)}) \right) \right)_{\tau=1}^{T^{(i)}} \subset E_i,$$

and a respective post-departure time-of-arrival process $\mathcal{T}^{(i)}$, defined using the partial sum of edge costs

$$\mathcal{T}^{(i)} := \left(\sum_{\gamma=1}^{\tau} c_i(\gamma, \theta_{\gamma-2}^{(i)}, \chi_{\gamma-2}^{(i)}, \theta_{\gamma-1}^{(i)}, \chi_{\gamma-1}^{(i)}) \right)_{\tau=1}^{T^{(i)}}.$$

Ideally, we would like to ensure that all edges in spatial conflict are non-overlapping in time. Before making this explicit, we must augment $C_{\mathcal{P}_{\cup N}}$ to account for flight levels. Indeed, we define

$$C_{\mathcal{P}_{\cup N}}^{H_{\text{legal}}} := \bigcup_{l=0}^{H} C_{\mathcal{P}_{\cup N}}^{(h_l)},$$

where we make explicit that $H_{\text{legal}} = \{h_0, h_1, \ldots, h_H\}$, and define the spatially conflicting, level-compatible sets $C_{\mathcal{P}_{\cup N}}^{(h)}$ for $h \in H_{\text{legal}}$ as

$$C_{\mathcal{P}_{\cup N}}^{(h)} := \left\{ \left(\left((\tau, \theta, \chi), (\tau{+}1, \theta', \chi') \right), \left((\tilde{\tau}, \tilde{\theta}, \tilde{\chi}), (\tilde{\tau}{+}1, \tilde{\theta}', \tilde{\chi}') \right) \right) \in C_{\mathcal{P}_{\cup N}} : \chi' {+} h_\tau^{(i)} = \tilde{\chi}' {+} h_{\tilde{\tau}}^{(j)} = h \right\}.$$

Now, we may clearly express the deconfliction condition as requiring

$$[\tau_0^{(i)} + \mathcal{T}_{\tau-1}^{(i)} - \Delta, \tau_0^{(i)} + \mathcal{T}_\tau^{(i)} + \Delta] \cap [\tau_0^{(j)} + \mathcal{T}_{\tilde{\tau}-1}^{(j)}, \tau_0^{(j)} + \mathcal{T}_{\tilde{\tau}}^{(j)}] = \emptyset \quad \forall (u_\tau^{(i)}, u_{\tilde{\tau}}^{(j)}) \in C_{\mathcal{P}_{\cup N}}^{H_{\text{legal}}},$$

where we adopt the convention that $\mathcal{T}_0^{(i)} := 0$ for all i, add the starting times $\tau_0^{(i)}, \tau_0^{(j)}$ to put all the subintervals on a common time unit, and note the time-buffer Δ (for pointwise separation this is often nonzero). So, to summarise, there are multiple ways in which a controller might mitigate conflicts:

- They may pick edges such that the resulting pairings are spatially deconflicted (so that resulting edges occupy $C^{\mathfrak{c}}_{\mathcal{P}_{\cup N}}$);
- They may pick differing flight levels to avoid spatial conflicts (so that the resulting edge pairs might occupy $C_{\mathcal{P}_{\cup N}}$ but not $C^{H_{\text{legal}}}_{\mathcal{P}_{\cup N}}$;
- They might pick well-timed edges and flight levels to avoid simultaneous occupation of the spatially conflicting, same-level subintervals (so that the resulting edge pairs might occupy $C^{H_{\text{legal}}}_{\mathcal{P}_{\cup N}}$, but the occupation interval sequences satisfy the Δ-buffered exclusive occupation condition above).

2.5 A Master Problem

We recall the original setup of the problem, and introduce all constraints. Supposing that there is a perturbable set of aircraft, given by $\mathcal{P}_{\cup N}$, the controller wishes to

$$\text{minimise } \{(\theta_\tau^{(i)},\chi_\tau^{(i)})_{\tau=1}^{T^{(i)}-1}\}_{i\in\mathcal{P}_{\cup N}} \mapsto \sum_{i\in\mathcal{P}_{\cup N}} \sum_{\tau=1}^{T^{(i)}-1} \left(2|\theta_\tau^{(i)} - \theta_{\tau-1}^{(i)}|^2 + |\chi_\tau^{(i)} - \chi_{\tau-1}^{(i)}|^2\right), \tag{2}$$

$$\begin{aligned}
\text{subject to} \quad & \theta_\tau^{(i)} \in \{-K, -K+1, \dots, K-1, K\} \quad \forall i, \tau, \\
& \chi_\tau^{(i)} \in \{-J, -J+1, \dots, J-1, J\} \quad \forall i, \tau, \\
& (\chi_\tau^{(i)} + h_\tau^{(i)}) \in H_{\text{legal}} \quad \forall i, \tau, && (3) \\
& |\tau_0^{(i)} + \mathcal{T}_{T^{(i)}} - \tau_1^{(i)}| \leq \mathcal{T} \quad \forall i, && (4) \\
& [\tau_0^{(i)} + \mathcal{T}_{\tau-1}^{(i)} - \Delta, \tau_0^{(i)} + \mathcal{T}_\tau^{(i)} + \Delta] \cap [\tau_0^{(j)} + \mathcal{T}_{\tilde{\tau}-1}^{(j)}, \tau_0^{(j)} + \mathcal{T}_{\tilde{\tau}}^{(j)}] = \emptyset \\
& \quad \forall (u_\tau^{(i)}, u_{\tilde{\tau}}^{(j)}) \in C^{H_{\text{legal}}}_{\mathcal{P}_{\cup N}}, && (5)
\end{aligned}$$

where:

- The primary objective (2) reflects the fact that we wish to issue the simplest instructions possible, with flight-level changes more desirable than lateral deviations, as discussed;
- Condition (3) reflects the fact that a flight level change might not always result in an operationally admissible flight level;
- The scheduling constraint (4) makes explicit the soft minimax formulation of our procedure, in which we wish to offer an upper bound on delays, as discussed;
- The final condition (5) ensures that any level-compatible spatial conflicts are deconflicted temporally.

Extending further, we now introduce an outer loop to the procedure in order to address controlling the size of $\mathcal{P}_{\cup N}$. In order to do this, we will need a natural heuristic for augmenting the perturbation set $\mathcal{P}_{\cup N}$, rather than optimising this as well (which would require simultaneously considering all deviation skeleton structures). This results in a greedy selection method, but certainly more effective than a uniformly random sampling rule. We build an understanding of the initial spatial conflict profile by defining, for each $j \in [N-1]$,

$$C^{(j)}_{\xi^{(N)}} := \{(u,v) \in E_N^\xi \times E_j^\xi : \underline{\mathrm{d}}(u,v) \leq \alpha, \}$$

then decomposing into a level-compatible union via

$$C^{(j,H_{\text{legal}})}_{\xi^{(N)}} := \bigcup_{h=0}^{H} C^{(j,h_l)}_{\xi^{(N)}},$$

where

$$C^{(j,h)}_{\xi^{(N)}} := \left\{ \left(\Big((\tau,\theta,\chi),(\tau{+}1,\theta',\chi')\Big), \Big((\tilde{\tau},\tilde{\theta},\tilde{\chi}),(\tilde{\tau}{+}1,\tilde{\theta}',\tilde{\chi}')\Big) \right) \in C^{(j)}_{\xi^{(N)}} : \chi'{+}h^{(N)}_{\tau} = \tilde{\chi}'{+}h^{(j)}_{\tilde{\tau}} = h \right\}.$$

One way of quantifying the level of confliction with incoming flight $\xi^{(N)}$ is to define spatiotemporal-violation waypoint-pairs

$$\mathcal{O}^{(j)}_{\xi^{(N)}} := \{(\tau,\tilde{\tau}) \in [T^{(N)}] \times [T^{(j)}] : (u^{(N)}_{\tau}, u^{(j)}_{\tilde{\tau}}) \in C^{(j)}_{\xi^{(N)}}\}.$$

Taking the sizes of this set over different j gives an understanding of the extent of confliction between incoming flight i and current flights. It is not decision-theoretically unnatural to consider perturbing current flights that are heavily in conflict - the resulting deviations could lead to a shared compromise between flight N and others, rather than excessive deviations for the incoming flight. Another desirable feature of this computation is that it can be done beforehand, without the combinatorially expensive contributions of the deviation skeletons.

To this end, we introduce the outer loop of the master problem, which incorporates the augmentation procedure, as well as a stopping criterion, which we discuss below.

1. Initialise the empty perturbation set $\mathcal{P} = \emptyset$ (the case where we simply attempt to slot in the incoming flight N).

2. Solve the minimisation problem (2) above with the initial perturbation set $\mathcal{P} = \emptyset$. This results in a complexity (primary objective) of $\mathcal{C}(\emptyset)$ (defined as ∞ if the solution is infeasible);

3. Choose a new candidate j^* for perturbation, where

$$j^* := \underset{j \in [N-1] \setminus \mathcal{P}}{\arg\max} \left\{ \left\| \mathcal{O}^{(j)}_{\xi^{(N)}} \right\| \right\},$$

   and add this element to $\mathcal{P}$. That is, $\mathcal{P} \leftarrow \mathcal{P} \cup \{j^*\}$.

4. Solve the minimisation problem (2) above with the new perturbation set $\mathcal{P}$, resulting in a complexity of $\mathcal{C}_{\text{new}}(\mathcal{P})$.

5. If $\mathcal{C}_{\text{new}} \geq \mathcal{C}$ (where $\mathcal{C}$ denotes the complexity from the prior perturbation set $\mathcal{P} \setminus \{j^*\}$), terminate the procedure due to an increase in instruction complexity. Otherwise, set $\mathcal{C}(\mathcal{P}) \leftarrow \mathcal{C}_{\text{new}}(\mathcal{P})$, and return to step 3.

Thanks to the atomicity of the deviation set, our termination procedure will not result in unnecessarily small perturbations of many flights. The total complexity will be bounded from above by $\mathcal{C}(\emptyset)$, which can be matched in complexity by a single flight-level change across $\mathcal{C}(\emptyset)/2$ flights (since deviating vertically by ± 1 requires an eventual return to the trajectory's vertical profile), which provides an upper bound on the size of $\mathcal{P}$ (on the assumption that it is finite, otherwise we can bound from above by half of the smallest-size $\mathcal{P}$ resulting in a finite complexity). By requiring a strictly monotone decrease in complexity, this upper bound cannot be achieved.

In cases where the final perturbation set $\mathcal{P}$ is too large, it is operationally arguable that the optimal procedure would rather be a ground-holding delay. As stated, this offloads operational

complexity onto the airport's optimisation of the departure queue, which we recognise as a last-resort that is outside of the scope of this work. Alternatively, one could work in an emergency procedure, where the framework is fed a new alternative route, for which deconfliction is repeated - this is a useful extension of our work that we will discuss later. In any case, the results of our procedure are interpretable enough that air traffic controllers may remain in-the-loop and activate other fall-back systems if necessary.

3 QUBO Formulation

Recalling the original fixed-$\mathcal{P}_{\cup N}$ master problem (2), we see ambiguity in the binary decision variable to eventually formulate: on one hand, all of the constraints can be understood in terms of instantaneous deviation decisions $(\theta_\tau^{(i)}, \chi_\tau^{(i)})$, but constraint (4) requires introduction of a slack variable and subsequent squaring for a valid penalty, which would result in a quartic. We linearise by augmenting the decision variable to an edge-selection variable, so that the problem sits on the level of the perturbation edge-sets $(E_i)_{i\in\mathcal{P}_{\cup N}}, (E_i^\xi)_{i\in\mathcal{P}_{\cup N}^c}$ introduced in Section 2.4. To this end, we begin by appropriately one-hot encoding the edge choices.

We do so lexicographically: for each waypoint $\tau - 1$ to τ edge of a general flight, we order according to the following priority alphabet, which has been chosen specifically for easily handling condition (3):

1. $\chi_{\tau-1} \in \{-J, -J+1, \dots, J-1, J\}$ (the outgoing flight level deviation;
2. $\theta_{\tau-1} \in \{-K, -K+1, \dots, K-1, K\}$ (the outgoing lateral deviation);
3. $\chi_{\tau-2} \in \{-J, -J+1, \dots, J-1, J\}$ (the incoming flight level deviation);
4. $\theta_{\tau-2} \in \{-K, -K+1, \dots, K-1, K\}$ (the incoming lateral deviation).

Using this lexicographic ordering, the problem can be converted into a binary selection problem, where one chooses edges $e_1^{(i)}, e_{T^{(i)}}^{(i)} \in \mathbb{B}((2J+1)(2K+1)), e_2^{(i)}, \dots, e_{T^{(i)}-1}^{(i)} \in \mathbb{B}((2J+1)^2(2K+1)^2)$, recalling that the first and last deviations are fixed to nominal. Stacking these objects, we denote the full path (whose linkages we will impose) of flight i and subsequently that of all perturbed flights by

$$e^{(i)} := \begin{bmatrix} e_1^{(i)} \\ e_2^{(i)} \\ \vdots \\ e_{T^{(i)}-1}^{(i)} \\ e_{T^{(i)}}^{(i)} \end{bmatrix}, \quad e_{\mathcal{P}_{\cup N}} := \begin{bmatrix} e^{(i_1)} \\ e^{(i_2)} \\ \vdots \end{bmatrix}, \quad i_1, i_2, \dots \in \mathcal{P}_{\cup N}.$$

This results in a binary vector $e_{\mathcal{P}_{\cup N}}$ of length $\sum_{l:i_l\in\mathcal{P}_{\cup N}}((T^{(i)}-2)(2J+1)^2(2K+1)^2 + 2(2J+1)(2K+1))$.

3.1 Hard Constraints

We begin the hard constraint QUBO formulations with the one-hot encoding of the full decision vector $e_{\mathcal{P}_{\cup N}}$. Denoting $M := (2J+1)^2(2K+1)^2$ (and, abusing notation slightly, $M := (2J+$

$1)(2K+1)$ for the endpoint edges), we wish to penalise

$$\begin{aligned}\sum_{i\in\mathcal{P}_{\cup N}}\sum_{\tau=1}^{T^{(i)}}\left(1-\mathbf{1}_M^\top e_\tau^{(i)}\right)^2 &= \sum_{i\in\mathcal{P}_{\cup N}} T^{(i)} + \sum_{i\in\mathcal{P}_{\cup N}}\sum_{\tau=1}^{T^{(i)}}\left((e_\tau^{(i)})^\top \mathbf{1}_M\mathbf{1}_M^\top e_\tau^{(i)} - 2(e_\tau^{(i)})^\top\mathbf{1}_M\right)\\ &= \sum_{i\in\mathcal{P}_{\cup N}} T^{(i)} + \sum_{i\in\mathcal{P}_{\cup N}}\sum_{\tau=1}^{T^{(i)}}(e_\tau^{(i)})^\top\left(\mathbf{1}_M\mathbf{1}_M^\top - 2\mathbb{I}_M\right)e_\tau^{(i)}\\ &= \sum_{i\in\mathcal{P}_{\cup N}} T^{(i)} + e_{\mathcal{P}_{\cup N}}^\top\left(\mathbb{I}_{(\sum_{i\in\mathcal{P}_{\cup N}}T^{(i)})}\otimes\left(\mathbf{1}_M\mathbf{1}_M^\top - 2\mathbb{I}_M\right)\right)e_{\mathcal{P}_{\cup N}},\end{aligned}$$

so the one-hot encoding QUBO matrix is defined as $H_{\text{one-hot}} := \mathbb{I}_{(\sum_{i\in\mathcal{P}_{\cup N}}T^{(i)})}\otimes\left(\mathbf{1}_M\mathbf{1}_M^\top - 2\mathbb{I}_M\right)$. We now proceed to reformulate the primary objective (2). Noting that they are lexicographically different in priority, we define elementary pricing blocks,

$$\Theta(\theta) := \mathbf{1}_{2J+1}\bigotimes\begin{bmatrix}(-K-\theta)^2\\(-K+1-\theta)^2\\\vdots\\(K-1-\theta)^2\\(K-\theta)^2\end{bmatrix},\quad \mathcal{F}(\chi) := \begin{bmatrix}(-J-\chi)^2\\(-J+1-\chi)^2\\\vdots\\(J-1-\chi)^2\\(J-\chi)^2\end{bmatrix}\bigotimes\mathbf{1}_{2K+1},$$

so that the primary objective QUBO matrices for given flight i and intermediate waypoint edges τ are given by

$$\Theta^{(i,\tau)} := \begin{cases}\mathbb{I}_{2J+1}\otimes\operatorname{diag}\{\Theta(\theta)\}_{\theta=-K}^{K}, & \tau = 2,\dots,T^{(i)}-1,\\ \operatorname{diag}\{\Theta(\theta)\}_{\theta=-K}^{K}, & \tau = 1, T^{(i)},\end{cases} \tag{6}$$

$$\mathcal{F}^{(i,\tau)} := \begin{cases}\operatorname{diag}\{\mathbf{1}_{2K+1}\otimes\mathcal{F}(\chi)\}_{\chi=-J}^{J}, & \tau = 2,\dots,T^{(i)}-1,\\ \operatorname{diag}\{\mathcal{F}(\chi)\}_{\chi=-J}^{J}, & \tau = 1, T^{(i)},\end{cases} \tag{7}$$

which cleanly provide the full primary objective QUBO matrices as

$$\begin{aligned}\Theta_{\mathcal{P}_{\cup N}} &:= 2\text{Blockdiag}\{\text{Blockdiag}\{\Theta^{(i,\tau)}\}_{\tau=1}^{T^{(i)}}\}_{i\in\mathcal{P}_{\cup N}},\\ \mathcal{F}_{\mathcal{P}_{\cup N}} &:= \text{Blockdiag}\{\text{Blockdiag}\{\mathcal{F}^{(i,\tau)}\}_{\tau=1}^{T^{(i)}}\}_{i\in\mathcal{P}_{\cup N}}.\end{aligned}$$

Forming the matrix H_{legal}-adherence QUBO matrix from (3), which we will denote the same, follows a similar procedure. Assuming that step changes for the next waypoint are promptly completed following the arrival at the previous waypoint, we need only account for the first lexicographic object in each edge - that is, the target vertical deviation. Indeed, more simply denoting vector stacking with notation $\oplus$, we define illegal level indicator vectors associated to waypoint edges τ as

$$\mathbb{1}^{(i,\tau)} := \begin{cases}\left(\bigoplus_{j=-J}^{J}\mathbb{1}\{(j+h_\tau^{(i)}\notin H_{\text{legal}}\}\right)\bigotimes\mathbf{1}_{(2K+1)} & \tau = 1,\\ \left(\bigoplus_{j=-J}^{J}\mathbb{1}\{(j+h_\tau^{(i)}\notin H_{\text{legal}}\}\right)\bigotimes\mathbf{1}_{(2J+1)(2K+1)^2} & \tau = 2,\dots,T^{(i)}-1,\\ \mathbf{0}_{(2J+1)(2K+1)} & \tau = T^{(i)},\end{cases}$$

whose entries are active for vertical deviations resulting in non-permitted flight levels, extended accordingly to account for the full lexicon. We can then extend this to the full QUBO matrix over all flights and waypoints, defined as

$$H_{\text{legal}} := \text{Blockdiag}\left\{\operatorname{diag}\{\mathbb{1}^{(i,\tau)}\}_{\tau=1}^{T^{(i)}}\right\}_{i\in\mathcal{P}_{\cup N}}.$$

Due to our deliberate lexicographic ordering, this will result in a sparse matrix, whose entries can be filled in by inserting blocks of ones for upper and lower violations - under any other ordering, this filling process would be significantly more involved.

In the following proposition we present the linkage QUBO matrix, whose form is nontrivial, owing to employing lexicographic ordering over augmented tensors.

Proposition 3. *Suppose that for flight i, at waypoint τ we select edge $e_\tau^{(i)}$, followed by $e_{\tau+1}^{(i)}$ at waypoint $\tau+1$, corresponding to deviation choices (a,b,c,d) and (e,f,g,h) respectively. Then the linkage constraint matrices $L_\tau^{(i)}$, defined using basis vectors $\{e_m\}_m \in \mathbb{B}((2J+1)(2K+1))$ as*

$$L_\tau^{(i)} := \begin{cases} \sum_{m=1}^{(2J+1)(2K+1)} e_m \big(\mathbf{1}_{(2J+1)(2K+1)} \otimes e_m\big)^\top, & \tau = 1, \\ \sum_{m=1}^{(2J+1)(2K+1)} \big(e_m \otimes \mathbf{1}_{(2J+1)(2K+1)}\big)\big(\mathbf{1}_{(2J+1)(2K+1)} \otimes e_m\big)^\top, & \tau = 2,\dots,T^{(i)}-2, \\ \sum_{m=1}^{(2J+1)(2K+1)} \big(e_m \otimes \mathbf{1}_{(2J+1)(2K+1)}\big) e_m^\top, & \tau = T^{(i)}-1 \end{cases}$$

have the property that

$$(e_\tau^{(i)})^\top L_\tau^{(i)} e_{\tau+1}^{(i)} = \begin{cases} 1 & \text{if } (a,b) = (g,h), \\ 0 & \text{otherwise.} \end{cases}$$

That is, the $L_\tau^{(i)}$ are the appropriate link compatibility matrices for our lexicon.

Proof. We begin by defining the lexicon coordinate map in the case of vertex selection, denoted by $\phi : \{-J, -J+1, \dots, J-1, J\} \times \{-K, -K+1, \dots, K-1, K\}$, where

$$\phi(\chi,\theta) := 1 + (\chi + J)(2K+1) + (\theta + K) \in \{1, \dots, (2J+1)(2K+1)\}$$

provides the position in the vector flattened according to the lexicon. The full edge lexicon map is then given by

$$\ell(\chi_a, \theta_a, \chi_b, \theta_b) := (2J+1)(2K+1)(\phi(\chi_a,\theta_a) - 1) + \phi(\chi_b,\theta_b),$$

so that

$$e_\tau^{(i)} = e_{\ell(a,b,c,d)}, \quad e_{\tau+1}^{(i)} = e_{\ell(e,f,g,h)},$$

where the e_ℓ are the basis vectors, instead of length $(2J+1)^2(2K+1)^2$. Using the fact that $e_{\ell(a,b,c,d)} = e_{\phi(a,b)} \otimes e_{\phi(c,d)}$ and accordingly for (e,f,g,h), we explicitly verify

$$\begin{aligned} &(e_\tau^{(i)})^\top L_\tau^{(i)} e_{\tau+1}^{(i)} = e_{\ell(a,b,c,d)}^\top L_\tau^{(i)} e_{\ell(e,f,g,h)} \\ &= \big(e_{\phi(a,b)} \otimes e_{\phi(c,d)}\big)^\top \left(\sum_{m=1}^{(2J+1)(2K+1)} \big(e_m \otimes \mathbf{1}_{(2J+1)(2K+1)}\big)\big(\mathbf{1}_{(2J+1)(2K+1)} \otimes e_m\big)^\top \right) \big(e_{\phi(e,f)} \otimes e_{\phi(g,h)}\big) \\ &= \sum_{m=1}^{5(2K+1)} \delta_{\phi(a,b),m}\delta_{m,\phi(g,h)} = \delta_{\phi(a,b),\phi(g,h)}. \end{aligned}$$

The case for the first and final linkages follows accordingly. □

To finally construct the full QUBO matrix for edge linkage, for each flight i define

$$B_\tau^{(i)} := \big(\mathbf{1}_M \mathbf{1}_M^\top - L_\tau^{(i)}\big), \quad \tau = 1, \dots, T^{(i)} - 1,$$

where $L_\tau^{(i)}$ is the linking matrix between the edge blocks at waypoints τ and $\tau+1$, and we recall that M depends on the waypoint in question, as for the one-hot QUBO matrix $H_{\text{one-hot}}$. The

linking contribution associated to flight i may then be written in upper-triangular form as

$$H_{\text{link}}^{(i)} = \begin{bmatrix} \mathbb{O} & B_1^{(i)} & \mathbb{O} & \cdots & \mathbb{O} \\ \mathbb{O} & \mathbb{O} & B_2^{(i)} & \ddots & \vdots \\ \mathbb{O} & \mathbb{O} & \mathbb{O} & \ddots & \mathbb{O} \\ \vdots & \ddots & \ddots & \ddots & B_{T^{(i)}-2}^{(i)} \\ \mathbb{O} & \cdots & \mathbb{O} & \mathbb{O} & \mathbb{O} \end{bmatrix}.$$

The full realised edge-linking matrix is then given by

$$H_{\text{link}} := \text{Blockdiag}\{H_{\text{link}}^{(i)}\}_{i \in \mathcal{P} \cup \mathcal{N}}.$$

3.2 Prompt Arrival

For constraints (4), (5), we will need to make use of the transition costs computed in Section 2.2. Indeed, we define the τ-stopped cost vector $C_{(\tau)}^{(i)}$ for flight i as

$$C_{(\tau)}^{(i)} := \big(\bigoplus_{l=1}^{\tau} C^{(i,\tau)}\big) \oplus \big(\bigoplus_{l=\tau+1}^{T^{(i)}} \mathbf{0}_M\big),$$

where each $C^{(i,\tau)}$ denotes the lexicographically flattened costs associated to traversing from waypoint $\tau - 1$ to τ. We simplify the full cost vector notation as $C^{(i)} \equiv C_{T^{(i)}}^{(i)}$. With this convention, we can rewrite the scheduling condition (4) in terms of the decision vector $e^{(i)}$ as

$$|\tau_0^{(i)} + \mathcal{T}_{T^{(i)}} - \tau_1^{(i)}| = |\tau_0^{(i)} + (C^{(i)})^\top e^{(i)} - \tau_1^{(i)}| \leq \mathcal{T}.$$

Adding slack variables and squaring, we see immediately that the cross terms arising from the outer product $(C^{(i)})(C^{(i)})^\top$ can be potentially large. Much like in [6], we will see computational benefit from formulating the problem as choosing optimal delay structures relative to the nominal one, rather than naively traversing literal edge costs. Indeed, we define the nominal costs vector as $\mathcal{C}^{(i)} := \bigoplus_{\tau=1}^{T^{(i)}} \mathcal{C}^{(i,\tau)}$, where each $\mathcal{C}^{(i,\tau)}$ is a scalar object. We also define the deviation-augmented nominal costs vector as

$$\widetilde{\mathcal{C}}^{(i)} := \bigoplus_{\tau=1}^{T^{(i)}} \big(\mathbf{1}_M \otimes \mathcal{C}^{(i,\tau)}\big),$$

and the associated τ-stopped vectors $\mathcal{C}_{(\tau)}^{(i)}, \widetilde{\mathcal{C}}_{(\tau)}^{(i)}$ accordingly as for $C_{(\tau)}^{(i)}$. Doing this, we note that the condition can be rewritten as an accumulated-delays penalty,

$$\begin{aligned} \mathcal{T} &\geq |\tau_0^{(i)} + (C^{(i)})^\top e^{(i)} - \tau_1^{(i)}| = |\tau_0^{(i)} + \mathcal{C}^{(i)} + (C^{(i)} - \widetilde{\mathcal{C}}^{(i)})^\top e^{(i)} - \tau_1^{(i)}| \\ &= |(C^{(i)} - \mathcal{C}^{(i)})^\top e^{(i)}| =: |(\Delta C^{(i)})^\top e^{(i)}|, \end{aligned}$$

where we have defined the full delays vector $\Delta C^{(i)}$. It is clear that the outer product of the full delays vector will be of a significantly reduced scale compared to that of the naive edge cost formulation.

With this reformulation, we introduce slack variables and decompose them into auxiliary decision variables $s_-^{(i)}, s_+^{(i)}$ and binary encoding vectors $B_-^{(i)}, B_+^{(i)}$ respectively, so that the inequalities become equalities given by

$$\begin{aligned} 0 &= (\Delta C^{(i)})^\top e^{(i)} - \mathcal{T} + (B_-^{(i)})^\top s_-^{(i)}, \\ 0 &= (\Delta C^{(i)})^\top e^{(i)} + \mathcal{T} - (B_+^{(i)})^\top s_+^{(i)}, \end{aligned}$$

where the binary encoding and auxiliary decision vectors are large enough to accommodate the slack range $[0, 2\mathcal{T}]$. Squaring and expanding, one can form the QUBO matrix associated to the scheduling constraint, denoted as $H_{\mathcal{T}}$. Note that we will introduce further auxiliary variables in the next section, so we will appropriately embed all previous QUBO matrices $H_{\text{one-hot}}, \Theta_{\mathcal{P}_{\cup N}}, \mathcal{F}_{\mathcal{P}_{\cup N}}, H_{\text{legal}}, H_{\text{link}}$ and $H_{\mathcal{T}}$ into their fully augmented versions once the temporal conflict penalties are formulated.

3.3 Temporal Deconfliction

Managing the deconfliction condition (5) is significantly more involved, owing to the fact that many combinations of deviations (and therefore trajectories) can lead to edge pairs in $C^{H_{\text{legal}}}_{\mathcal{P}_{\cup N}}$. Using DP-based methods for forward and backward information propagation based on available and remaining time budgets, we are able to efficiently classify temporal conflict cases between subintervals. Following this, we introduce a disjunctive programming framework to handle these conflict cases in QUBO form, and use our precomputed time budgets to tighten constants and slack ranges for the involved equations. As a final reduction of complexity, we reformulate the QUBO in terms of delays rather than cumulative times in order to prevent buildup of the outer product term contributing to the final QUBO. Albeit being theoretically involved, the practical implementation is highly parallel, and is certainly more computationally efficient than constructing temporal penalties for any edge pair in $C^{H_{\text{legal}}}_{\mathcal{P}_{\cup N}}$ - after all, if two flights are level-compatible and spatially conflicting, but an hour apart, it is a waste to penalise their conflicts.

Recalling now our level-compatible spatial conflict set, $C^{H_{\text{legal}}}_{\mathcal{P}_{\cup N}}$, and that we were able to navigate across the relevant edge sets E_i via waypoint-deviation coordinates, we augment this coordinate system to flight-waypoint-deviation coordinates, in order to navigate across the full conflicts set. In fact, we also now include the subinterval index of the relevant edge, as this provides a fixed offset for each deconfliction condition. Precisely, we suppose that the conflicts set can be enumerated as

$$C^{H_{\text{legal}}}_{\mathcal{P}_{\cup N}} = \{\kappa_l\}_{l=1}^{\text{PP}} \cup \{\sigma_l\}_{l=1}^{\text{PF}},$$

where each $\{\kappa_l\}_{l=1}^{\text{PP}}$ denotes the set of spatially conflicting, level-compatible edge pairs between pairs of perturbable flights, and $\{\sigma_l\}_{l=1}^{\text{PF}}$ that between perturbable and non-perturbable flight pairs. Each edge pair is of the form

$$\kappa_l = \Big(\overbrace{(\kappa^{(1)}_{l,1}, \kappa^{(2)}_{l,1}, \kappa^{(3)}_{l,1}, \kappa^{(4)}_{l,1})}^{\text{perturbable flight \#1}}, \overbrace{(\kappa^{(1)}_{l,2}, \kappa^{(2)}_{l,2}, \kappa^{(3)}_{l,2}, \kappa^{(4)}_{l,2})}^{\text{perturbable flight \#2}} \Big), \quad \sigma_l = \Big(\overbrace{(\sigma^{(1)}_{l,1}, \sigma^{(2)}_{l,1}, \sigma^{(3)}_{l,1}, \sigma^{(4)}_{l,1})}^{\text{perturbable flight}}, \overbrace{(\sigma^{(1)}_{l,2}, \sigma^{(2)}_{l,2}, \sigma^{(3)}_{l,2})}^{\text{non-perturbable flight}} \Big).$$

$\kappa^{(1)}_{l,1} :=$ flight number of first edge
$\kappa^{(2)}_{l,1} :=$ waypoint index of first edge
$\kappa^{(3)}_{l,1} :=$ subinterval index of first edge
$\kappa^{(4)}_{l,1} :=$ lateral-vertical deviations of first edge
$\kappa^{(1)}_{l,2} :=$ flight number of last edge-pair
$\kappa^{(2)}_{l,2} :=$ waypoint index of last edge
$\kappa^{(3)}_{l,2} :=$ subinterval index of last edge
$\kappa^{(4)}_{l,2} :=$ lateral-vertical deviation of last edge
$\sigma^{(1)}_{l,1} :=$ flight number of first edge
$\sigma^{(2)}_{l,1} :=$ waypoint index of first edge
$\sigma^{(3)}_{l,1} :=$ subinterval index of the first edge
$\sigma^{(4)}_{l,1} :=$ lateral-vertical deviation of first edge
$\sigma^{(1)}_{l,2} :=$ flight number of last edge
$\sigma^{(2)}_{l,2} :=$ waypoint index of last edge
$\sigma^{(3)}_{l,2} :=$ subinterval index of last edge

Being more precise now with our notation, we denote the amount of time it takes flight $\kappa^{(1)}$ to clear subinterval $\kappa^{(3)}$ of waypoint edge $\kappa^{(2)}$ corresponding to lateral-vertical deviation $\kappa^{(4)}$ (here we have simplified notation slightly in removing subscripts, as we are just considering one

flight) as

$$t^{(\kappa^{(1)})}(\kappa^{(2)},\kappa^{(3)},\kappa^{(4)}) = t^{(\kappa^{(1)})}(\kappa^{(2)}-1,\kappa^{(4)}) + \delta_{\text{offset}}(\kappa),$$

where $t^{(\kappa^{(1)})}(\kappa^{(2)}-1,\kappa^{(4)})$ denotes the time it takes to reach the start of waypoint edge $\kappa^{(2)}$ corresponding to lateral-vertical deviation $\kappa^{(4)}$, then shifted by $\delta_{\text{offset}}(\kappa)$, which denotes the time it takes to clear the given subinterval $\kappa^{(3)}$ along that full edge. For a nominal flight edge instance $\sigma_2 = (\sigma_2^{(1)},\sigma_2^{(2)},\sigma_2^{(3)})$ we denote this object as $t^{(\sigma_2^{(1)})}(\sigma_2^{(2)},\sigma_2^{(3)},\cdot)$, making explicit the lack of edge choice.

3.3.1 Temporal Conflict Classification

As alluded to above, we will need to consider the time envelope of possible arrival times, resulting from being able to manoeuvre freely before any waypoint. We denote the maximal time it takes flight $\kappa^{(1)}$ to clear subinterval $\kappa^{(3)}$ of waypoint edge $\kappa^{(2)}$ corresponding to lateral-vertical deviation $\kappa^{(4)}$ as

$$\overline{t}^{(\kappa^{(1)})}(\kappa^{(2)},\kappa^{(3)},\kappa^{(4)}) = \overline{t}^{(\kappa^{(1)})}(\kappa^{(2)}-1,\kappa^{(4)}) + \delta_{\text{offset}}(\kappa),$$

where $\overline{t}^{(\kappa^{(1)})}(\kappa^{(2)}-1,\kappa^{(4)})$ is defined accordingly; the lower bound of the time envelope is defined accordingly with notation $\underline{t}$. Both of these bounds can be computed efficiently using a dynamic programming procedure, as outlined in APPENDIX.

Splitting condition (5) into a pair of ordering constraints, for any conflicting pair $\kappa = (\kappa_1,\kappa_2) \in C_{\mathcal{P}_{\cup N}}^{H_{\text{legal}}}$, we would like to satisfy either

$$t^{(\kappa_1^{(1)})}(\kappa_1^{(2)},\kappa_1^{(3)},\kappa_1^{(4)}) + \Delta \le t^{(\kappa_2^{(1)})}(\kappa_2^{(2)},\kappa_2^{(3)}-1,\kappa_2^{(4)}) \tag{X}$$

$$\text{or} \quad t^{(\kappa_2^{(1)})}(\kappa_2^{(2)},\kappa_2^{(3)},\kappa_2^{(4)}) + \Delta \le t^{(\kappa_1^{(1)})}(\kappa_1^{(2)},\kappa_1^{(3)}-1,\kappa_1^{(4)}). \tag{Y}$$

In other words, one plane should exit at least Δ time units before the other enters the area of level-compatible, spatial conflict. To further inspect the temporal conflicts, we define time-envelope bounds

$$\begin{aligned}
a_X^{(\kappa)} &:= \underline{t}^{(\kappa_1^{(1)})}(\kappa_1^{(2)},\kappa_1^{(3)},\kappa_1^{(4)}) + \Delta - \overline{t}^{(\kappa_2^{(1)})}(\kappa_2^{(2)},\kappa_2^{(3)}-1,\kappa_2^{(4)}),\\
b_X^{(\kappa)} &:= \overline{t}^{(\kappa_1^{(1)})}(\kappa_1^{(2)},\kappa_1^{(3)},\kappa_1^{(4)}) + \Delta - \underline{t}^{(\kappa_2^{(1)})}(\kappa_2^{(2)},\kappa_2^{(3)}-1,\kappa_2^{(4)}),\\
a_Y^{(\kappa)} &:= \underline{t}^{(\kappa_2^{(1)})}(\kappa_2^{(2)},\kappa_2^{(3)},\kappa_2^{(4)}) + \Delta - \overline{t}^{(\kappa_1^{(1)})}(\kappa_1^{(2)},\kappa_1^{(3)}-1,\kappa_1^{(4)}),\\
b_Y^{(\kappa)} &:= \overline{t}^{(\kappa_2^{(1)})}(\kappa_2^{(2)},\kappa_2^{(3)},\kappa_2^{(4)}) + \Delta - \underline{t}^{(\kappa_1^{(1)})}(\kappa_1^{(2)},\kappa_1^{(3)}-1,\kappa_1^{(4)}),
\end{aligned}$$

so that

$$\begin{aligned}
t^{(\kappa_1^{(1)})}(\kappa_1^{(2)},\kappa_1^{(3)},\kappa_1^{(4)}) + \Delta - t^{(\kappa_2^{(1)})}(\kappa_2^{(2)},\kappa_2^{(3)}-1,\kappa_2^{(4)}) &\in [a_X^{(\kappa)}, b_X^{(\kappa)}],\\
t^{(\kappa_2^{(1)})}(\kappa_2^{(2)},\kappa_2^{(3)},\kappa_2^{(4)}) + \Delta - t^{(\kappa_1^{(1)})}(\kappa_1^{(2)},\kappa_1^{(3)}-1,\kappa_1^{(4)}) &\in [a_Y^{(\kappa)}, b_Y^{(\kappa)}].
\end{aligned}$$

The following provides us with a simple means of deducing whether the κ are temporally safe, which is the first temporal conflict case. Really, it confirms the standard intuition that if one ordering is inevitable, then the other ordering is rendered impossible.

Proposition 4. *With κ as above, if one ordering is guaranteed, then the other is impossible. That is,*

$$(b_X^{(\kappa)} < 0 \implies a_Y^{(\kappa)} > 0) \quad \textit{and} \quad (b_Y^{(\kappa)} < 0 \implies a_X^{(\kappa)} > 0)$$

Proof. We verify the first statement only. Writing explicitly,

$$\begin{aligned} a_Y^{(\kappa)} &= \underline{t}^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)}, \kappa_2^{(4)}) + \Delta - \overline{t}^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)} - 1, \kappa_1^{(4)}) \\ &= \underline{t}^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)} - 1, \kappa_2^{(4)}) + c_{\kappa_2^{(1)}}(\kappa_2^{(2)}, \kappa_2^{(3)}, \kappa_2^{(4)}) \\ &\quad + \Delta - \overline{t}^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)}, \kappa_1^{(4)}) + c_{\kappa_1^{(1)}}(\kappa_2^{(2)}, \kappa_2^{(3)}, \kappa_2^{(4)}) \\ &= c_{\kappa_2^{(1)}}(\kappa_2^{(2)}, \kappa_2^{(3)}, \kappa_2^{(4)}) + \Delta + c_{\kappa_1^{(1)}}(\kappa_2^{(2)}, \kappa_2^{(3)}, \kappa_2^{(4)}) - b_X^{(\kappa)} \end{aligned}$$

where we have adapted the edge cost notation from Section 2.4 slightly in order to accommodate subintervals too. By nonnegativity of the edge costs, the claim follows. □

We see from the above that all terms in $C^{H_{\text{legal}}}_{\mathcal{P}_{\cup N}}$ are deemed temporally deconflicted if either $b_X^{(\kappa)} < 0$ or $b_Y^{(\kappa)} < 0$ (the intersection is not possible). Conversely, a temporal conflict potentially exists if $b_X^{(\kappa)}, b_Y^{(\kappa)} > 0$, with the type of temporal conflict depending on the the signs of $a_X^{(\kappa)}, a_Y^{(\kappa)}$. We summarise these cases below in Table 1:

Table 1: Temporal conflict case classification in terms of envelopes $[a_X^{(\kappa)}, b_X^{(\kappa)}], [a_Y^{(\kappa)}, b_Y^{(\kappa)}]$.

Case	Conditions	Interpretation
Safe	$a_X^{(\kappa)} < b_X^{(\kappa)} < 0$ or $a_Y^{(\kappa)} < b_Y^{(\kappa)} < 0$	Ordering guaranteed, no temporal penalty required
Force (X) only	$a_X^{(\kappa)} < 0 < b_X^{(\kappa)},\ 0 < a_Y^{(\kappa)} < b_Y^{(\kappa)}$	Y ordering impossible, X is feasible
Force (Y) only	$0 < a_X^{(\kappa)} < b_X^{(\kappa)},\ a_Y^{(\kappa)} < 0 < b_Y^{(\kappa)}$	X ordering impossible, Y is feasible
Choice	$a_X^{(\kappa)} < 0 < b_X^{(\kappa)},\ a_Y^{(\kappa)} < 0 < b_Y^{(\kappa)}$	Both orderings feasible, binary choice variable needed.
Temporally impossible	$0 < a_X^{(\kappa)} < b_X^{(\kappa)},\ 0 < a_Y^{(\kappa)} < b_Y^{(\kappa)}$	Neither ordering feasible

On the other hand, this classification method, where we propagate information forward based on possible arrival times, will almost certainly permit deviated trajectories whose incurred delays cannot be compensated following the conflicting subinterval in question. Whilst this is technically fine due to the nested property of the conflict cases in Table (1), practical challenges may arise regarding slack ranges and involved constants during implementation, due to the widening intervals towards the ends of trajectories. Owing to Condition (4), we can actually significantly reduce the time windows towards the ends of trajectories by propagating backwards information about remaining time budgets. Namely, if we save more time than $\mathcal{T}$ up to the final waypoint, we may not have sufficient delay available in the remaining time budget to compensate and complete the flight within the required scheduling constraint. To this end, where we are concerned with some subinterval from coordinate set κ, we introduce the decomposition of the final realised flight time as

$$\mathcal{T}_{T^{(\kappa^{(1)})}} := t^{(\kappa^{(1)})}(\kappa^{(2)}, \kappa^{(3)}, \kappa^{(4)}) + r^{(\kappa^{(1)})}(\kappa^{(2)}, \kappa^{(3)}, \kappa^{(4)}),$$

where r denotes the remaining flight time, having cleared the subinterval detailed by κ. We can use this to rewrite the scheduling condition as

$$\tau_1^{(\kappa^{(1)})} - \tau_0^{(\kappa^{(1)})} - \mathcal{T} \leq t^{(\kappa^{(1)})}(\kappa^{(2)}, \kappa^{(3)}, \kappa^{(4)}) + r^{(\kappa^{(1)})}(\kappa^{(2)}, \kappa^{(3)}, \kappa^{(4)}) \leq \tau_1^{(\kappa^{(1)})} - \tau_0^{(\kappa^{(1)})} + \mathcal{T},$$

and introducing the equivalent notation $\overline{r}, \underline{r}$ for the residual times, we may deduce the time envelope condition

$$\begin{aligned} t^{(\kappa^{(1)})}(\kappa^{(2)}, \kappa^{(3)}, \kappa^{(4)}) \in [\tau_1^{(\kappa^{(1)})} - \tau_0^{(\kappa^{(1)})} - \mathcal{T} - \overline{r}^{(\kappa^{(1)})}(\kappa^{(2)}, \kappa^{(3)}, \kappa^{(4)}), \\ \tau_1^{(\kappa^{(1)})} - \tau_0^{(\kappa^{(1)})} + \mathcal{T} - \underline{r}^{(\kappa^{(1)})}(\kappa^{(2)}, \kappa^{(3)}, \kappa^{(4)}) \\ =: [\underline{t}_r^{(\kappa^{(1)})}(\kappa^{(2)}, \kappa^{(3)}, \kappa^{(4)}), \overline{t}_r^{(\kappa^{(1)})}(\kappa^{(2)}, \kappa^{(3)}, \kappa^{(4)})]. \end{aligned}$$

Combining these with the forward-propagated time intervals for general subinterval edge pair $\kappa = (\kappa_1, \kappa_2)$, we can deduce tighter, budget-aware time intervals, arising from the envelope bounds

$$\begin{aligned} \tilde{a}_X^{(\kappa)} &:= \max\left\{\underline{t}^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)}, \kappa_1^{(4)}), \underline{t}_r^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)}, \kappa_1^{(4)})\right\} + \Delta \\ &\quad - \min\left\{\overline{t}^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)} - 1, \kappa_2^{(4)}), \overline{t}_r^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)} - 1, \kappa_2^{(4)})\right\}, \\ \tilde{b}_X^{(\kappa)} &:= \min\left\{\overline{t}^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)}, \kappa_1^{(4)}), \overline{t}_r^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)}, \kappa_1^{(4)})\right\} + \Delta \\ &\quad - \max\left\{\underline{t}^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)} - 1, \kappa_2^{(4)}), \underline{t}_r^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)} - 1, \kappa_2^{(4)})\right\}, \\ \tilde{a}_Y^{(\kappa)} &:= \max\left\{\underline{t}^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)}, \kappa_2^{(4)}), \underline{t}_r^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)}, \kappa_2^{(4)})\right\} + \Delta \\ &\quad - \min\left\{\overline{t}^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)} - 1, \kappa_1^{(4)}), \overline{t}_r^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)} - 1, \kappa_1^{(4)})\right\}, \\ \tilde{b}_Y^{(\kappa)} &:= \min\left\{\overline{t}^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)}, \kappa_2^{(4)}), \overline{t}_r^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)}, \kappa_2^{(4)})\right\} + \Delta \\ &\quad - \max\left\{\underline{t}^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)} - 1, \kappa_1^{(4)}), \underline{t}_r^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)} - 1, \kappa_1^{(4)})\right\}. \end{aligned}$$

It is quick to check that the resulting time envelopes $[\tilde{a}_X^{(\kappa)}, \tilde{b}_X^{(\kappa)}] \subseteq [a_X^{(\kappa)}, b_X^{\kappa}], [\tilde{a}_Y^{(\kappa)}, \tilde{b}_Y^{(\kappa)}] \subseteq [a_Y^{(\kappa)}, b_Y^{(\kappa)}]$ only prune trajectories whose times cannot achieve the scheduling constraint (4).

One final simplification arises from an edge-level hierarchical collapsing rule for the cases. These result from an edge selection requiring simultaneous selection of all constituent subintervals. For instance, if one subinterval pairing is considered temporally impossible, this constraint dictates that both full associated edges become temporally impossible, regardless of what the other temporal conflict cases may be between the subinterval pairings. This is particularly helpful since the collapsing rules always result in simpler temporal conflict constraints. We present these logical statements concisely in Table 2, with events denoted as follows:

- $(\exists)T^c :=$ {(there exists a) temporally impossible (subinterval) edge};
- $(\exists)X :=$ {(there exists a) force X only (subinterval) edge};
- $(\exists)Y :=$ {(there exists a) force Y only (subinterval) edge}

Table 2: Hierarchical collapse rule for temporal subinterval pairings.

Condition	Collapse result
$\exists T^c$	T^c
$(\exists X) \cap (\exists Y)^c \cap (\exists T^c)^c$	X
$(\exists X)^c \cap (\exists Y) \cap (\exists T^c)^c$	Y
$(\exists X) \cap (\exists Y)$	T^c

3.3.2 Disjunctive Programming for Enforcing Deconfliction

We proceed by considering the case of a perturbable-perturbable pair $\kappa = (\kappa_1, \kappa_2)$, where the envelopes give a choice case. The other settings follow more simply, and we summarise them in Table 3.

Indeed, for the case of $\kappa \in C^{H_{\text{legal}}}_{\mathcal{P}_{\cup N}}$, we would like to enforce either (X) or (Y). This type of disjunctive programming is difficult to handle whilst maintaining second-order terms in the QUBO. For instance, we could lift to the space of possible paths, from which we could easily precompute orderings, but we would be exchanging the computational challenge of disjunctive programming for the combinatorially explosive route enumeration problem. In this work, we remain at the edge-selection level and settle for a big-M formulation, as presented in [7]. For our ordering inequalities, this would amount to introducing some ordering variable z_κ and enforcing both inequalities

$$t^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)}, \widetilde{\kappa}_1^{(4)}) + \Delta - t^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)} - 1, \widetilde{\kappa}_2^{(4)}) \leq M z_\kappa + M(2 - \mathbb{1}_{\kappa_1}^\top e^{(\kappa_1^{(1)})} - \mathbb{1}_{\kappa_2}^\top e^{(\kappa_2^{(1)})}),$$
$$t^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)}, \widetilde{\kappa}_2^{(4)}) + \Delta - t^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)} - 1, \widetilde{\kappa}_1^{(4)}) \leq M(1 - z_\kappa) + M(2 - \mathbb{1}_{\kappa_1}^\top e^{(\kappa_1^{(1)})} - \mathbb{1}_{\kappa_2}^\top e^{(\kappa_2^{(1)})}),$$

where $z_\kappa \in \mathbb{B}$ is a binary bit, and the vector $\mathbb{1}_{\kappa_i}$ denotes the vector of length equal to that of the decision vector for flight $\kappa_i^{(1)}$, with zeros everywhere, except for the entry corresponding to choosing lateral-vertical deviations $\kappa_i^{(4)}$ for waypoint edge $\kappa_i^{(2)}$ at subinterval $\kappa_i^{(3)}$. Note that the chosen deviation $\widetilde{\kappa}_i^{(4)}$ at the waypoint-subinterval index for each flight can now vary. For sufficiently large M, we see the following cases, leading to satisfying the original constraints (X),(Y):

- If only one or neither of the edges are selected, the second M term provides a large positive contribution to the inequality, regardless of the ordering variable. Given that all active flights in an airspace (the given context) will clearly enter and exit their respective waypoint edges within an interval (after all, flights cannot perpetually operate on finite fuel), we can be certain that these constraints will be met without having to enforce temporal deconfliction;

- In the event that both edges are selected, and that flight $\kappa_1^{(1)}$ reaches its subinterval before flight $\kappa_2^{(1)}$ reaches its own, we would like to enforce the inequality (X). The optimal choice will be to select $z_\kappa = 0$, so that the first inequality above collapses to (X) to enforce temporal deconfliction, and the second inequality collapses to a trivial one, so we don't enforce (Y). The converse ordering choice, $z_\kappa = 1$ would lead to unnecessary relaxation of the first inequality, and incorrectly collapsing the second to enforcing (Y), which could not yield zero penalty;

- The converse case follows equivalently.

Whilst the big-M formulation provides a means of approaching the disjunctive programming problem, M being too large will lead to extreme disparities across the dynamic range of the resulting QUBO, as well as requiring very large slack variables [1, 4]. Thankfully, the dynamic programming time envelopes, arising from the helpful feedforward structure of the edge graphs, will allow us to compute the tightest possible bounds on the M values, and subsequently the slack ranges required.

Once we decide to enforce such a constraint, matters are complicated slightly by the fact that there is no guarantee that deviations $\kappa_1^{(4)}$ and $\kappa_2^{(4)}$ are chosen. Rather, we must enforce the inequality over general $\widetilde{\kappa}_1^{(4)}, \widetilde{\kappa}_2^{(4)}$, which broaden the time envelopes we computed previously.

We compute the new time envelope bounds to be

$$\begin{aligned}
\alpha_X^{(\kappa)} &:= \min_{\widetilde{\kappa}_1^{(4)}} \left\{ \max\left\{ \underline{t}^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)}, \widetilde{\kappa}_1^{(4)}), \underline{t}_r^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)}, \widetilde{\kappa}_1^{(4)}) \right\} \right\} + \Delta \\
&\quad - \max_{\widetilde{\kappa}_2^{(4)}} \left\{ \min\left\{ \overline{t}^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)} - 1, \widetilde{\kappa}_2^{(4)}), \overline{t}_r^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)} - 1, \widetilde{\kappa}_2^{(4)}) \right\} \right\}, \\
\beta_X^{(\kappa)} &:= \max_{\widetilde{\kappa}_1^{(4)}} \left\{ \min\left\{ \overline{t}^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)}, \widetilde{\kappa}_1^{(4)}), \overline{t}_r^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)}, \widetilde{\kappa}_1^{(4)}) \right\} \right\} + \Delta \\
&\quad - \min_{\widetilde{\kappa}_2^{(4)}} \left\{ \max\left\{ \underline{t}^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)} - 1, \widetilde{\kappa}_2^{(4)}), \underline{t}_r^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)} - 1, \widetilde{\kappa}_2^{(4)}) \right\} \right\}, \\
\alpha_Y^{(\kappa)} &:= \min_{\widetilde{\kappa}_2^{(4)}} \left\{ \max\left\{ \underline{t}^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)}, \widetilde{\kappa}_2^{(4)}), \underline{t}_r^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)}, \widetilde{\kappa}_2^{(4)}) \right\} \right\} + \Delta \\
&\quad - \max_{\widetilde{\kappa}_1^{(4)}} \left\{ \min\left\{ \overline{t}^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)} - 1, \widetilde{\kappa}_1^{(4)}), \overline{t}_r^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)} - 1, \widetilde{\kappa}_1^{(4)}) \right\} \right\}, \\
\beta_Y^{(\kappa)} &:= \max_{\widetilde{\kappa}_2^{(4)}} \left\{ \min\left\{ \overline{t}^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)}, \widetilde{\kappa}_2^{(4)}), \overline{t}_r^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)}, \widetilde{\kappa}_2^{(4)}) \right\} \right\} + \Delta \\
&\quad - \min_{\widetilde{\kappa}_1^{(4)}} \left\{ \max\left\{ \underline{t}^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)} - 1, \widetilde{\kappa}_1^{(4)}), \underline{t}_r^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)} - 1, \widetilde{\kappa}_1^{(4)}) \right\} \right\},
\end{aligned}$$

so that $[\tilde{a}_X^{(\kappa)}, \tilde{b}_X^{(\kappa)}] \subseteq [\alpha_X^{(\kappa)}, \beta_X^{(\kappa)}], [\tilde{a}_Y^{(\kappa)}, \tilde{b}_Y^{(\kappa)}] \subseteq [\alpha_Y^{(\kappa)}, \beta_Y^{(\kappa)}]$.

With these, we can deduce the smallest M can feasibly be, by considering how positive the smallest possible activation of M must be for the largest possible instances of the time differences in the big-M constraints. Indeed, we separately define M for each constraint as

$$\mathcal{M}_X^{(\kappa)} := \beta_X^{(\kappa)}, \quad \mathcal{M}_Y^{(\kappa)} := \beta_Y^{(\kappa)},$$

and replace M in the big-M constraints accordingly. Then, for computing the slack ranges for each inequality, we take the highest possible activation of M and the most negative instances of the time differences in the big-M constraints, resulting in respective slack ranges

$$[0, 3\mathcal{M}_X^{(\kappa)} - \alpha_X^{(\kappa)}], \quad [0, 3\mathcal{M}_Y^{(\kappa)} - \alpha_Y^{(\kappa)}].$$

These limit the size of the auxiliary variables contributed, and the associated binary encoding vectors, resulting in the simplest possible constraints

$$\begin{aligned}
&t^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)}, \widetilde{\kappa}_1^{(4)}) + \Delta - t^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)} - 1, \widetilde{\kappa}_2^{(4)}) \\
&\quad - \mathcal{M}_X^{(\kappa)} z_\kappa - \mathcal{M}_X^{(\kappa)}(2 - \mathbb{1}_{\kappa_1}^\top e^{(\kappa_1^{(1)})} - \mathbb{1}_{\kappa_2}^\top e^{(\kappa_2^{(1)})}) + (B_-^{(\kappa)})^\top s_-^{(\kappa)} = 0, \\
&t^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)}, \widetilde{\kappa}_2^{(4)}) + \Delta - t^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)} - 1, \widetilde{\kappa}_1^{(4)}) \\
&\quad - \mathcal{M}_Y^{(\kappa)}(1 - z_\kappa) - \mathcal{M}_Y^{(\kappa)}(2 - \mathbb{1}_{\kappa_1}^\top e^{(\kappa_1^{(1)})} - \mathbb{1}_{\kappa_2}^\top e^{(\kappa_2^{(1)})}) + (B_+^{(\kappa)})^\top s_+^{(\kappa)} = 0,
\end{aligned}$$

where $s_-^{(\kappa)}, s_+^{(\kappa)}$ are the auxiliary variables, with the respective binary encoding vectors given by $B_-^{(\kappa)}, B_+^{(\kappa)}$. Note that the temporally impossible case is much simpler, resulting in a positive entry in the QUBO matrix where the position corresponds to the involved edge choices, made explicit in Table 3.

Just as in Section 3.2, we can gain considerable well-posedness of the problem by reformulating in terms of delay choices. In fact, it is even more critical in this instance, as full outer

products of the τ-stopped cost vectors $C^{(i)}_{(\tau)}$ are contributed for every temporal conflict past waypoint τ for flight i, in contrast to the auxiliary bits, which only ever receive one contribution each, corresponding to their unique associated conflicts. Noting, for instance, that we can rewrite

$$\begin{aligned} t^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)}, \widetilde{\kappa}_1^{(4)}) &= (C^{(\kappa_1^{(1)})}_{(\kappa_1^{(2)}-1)})^\top e^{(\kappa_1^{(1)})} + \delta_{\text{offset}}(\kappa_1) \\ &= \mathcal{C}^{(\kappa_1^{(1)})}_{(\kappa_1^{(2)}-1)} + \big(C^{(\kappa_1^{(1)})}_{(\kappa_1^{(2)}-1)} - \widetilde{\mathcal{C}}^{(\kappa_1^{(1)})}_{(\kappa_1^{(2)}-1)}\big)^\top e^{(\kappa_1^{(1)})} + \delta_{\text{offset}}(\kappa_1) \\ &= \mathcal{C}^{(\kappa_1^{(1)})}_{(\kappa_1^{(2)}-1)} + \big(\Delta C^{(\kappa_1^{(1)})}_{(\kappa_1^{(2)}-1)}\big)^\top e^{(\kappa_1^{(1)})} + \delta_{\text{offset}}(\kappa_1), \end{aligned}$$

we see that the final equality constraints may be rewritten as

$$\begin{aligned} 0 = {} & \mathcal{C}^{(\kappa_1^{(1)})}_{(\kappa_1^{(2)}-1)} + \big(\Delta C^{(\kappa_1^{(1)})}_{(\kappa_1^{(2)}-1)}\big)^\top e^{(\kappa_1^{(1)})} + \delta_{\text{offset}}(\kappa_1) + \Delta \\ & - \mathcal{C}^{(\kappa_2^{(1)})}_{(\kappa_2^{(2)}-1)} - \big(\Delta C^{(\kappa_2^{(1)})}_{(\kappa_2^{(2)}-1)}\big)^\top e^{(\kappa_2^{(1)})} - \delta_{\text{offset}}((\kappa_2^{(1)}, \kappa_2^{(2)}, \kappa_2^{(3)} - 1, \kappa_2^{(4)})) \\ & - \mathcal{M}_X^{(\kappa)} z_\kappa - \mathcal{M}_X^{(\kappa)}(2 - \mathbb{1}_{\kappa_1}^\top e^{(\kappa_1^{(1)})} - \mathbb{1}_{\kappa_2}^\top e^{(\kappa_2^{(1)})}) + (B_-^{(\kappa)})^\top s_-^{(\kappa)}, \\ 0 = {} & \mathcal{C}^{(\kappa_2^{(1)})}_{(\kappa_2^{(2)}-1)} + \big(\Delta C^{(\kappa_2^{(1)})}_{(\kappa_2^{(2)}-1)}\big)^\top e^{(\kappa_2^{(1)})} + \delta_{\text{offset}}(\kappa_2) + \Delta \\ & - \mathcal{C}^{(\kappa_1^{(1)})}_{(\kappa_1^{(2)}-1)} - \big(\Delta C^{(\kappa_1^{(1)})}_{(\kappa_1^{(2)}-1)}\big)^\top e^{(\kappa_1^{(1)})} - \delta_{\text{offset}}((\kappa_1^{(1)}, \kappa_1^{(2)}, \kappa_1^{(3)} - 1, \kappa_1^{(4)})) \\ & - \mathcal{M}_Y^{(\kappa)}(1 - z_\kappa) - \mathcal{M}_Y^{(\kappa)}(2 - \mathbb{1}_{\kappa_1}^\top e^{(\kappa_1^{(1)})} - \mathbb{1}_{\kappa_2}^\top e^{(\kappa_2^{(1)})}) + (B_+^{(\kappa)})^\top s_+^{(\kappa)}. \end{aligned}$$

Squaring and considering the resulting terms, one can see here that this reformulation trades off extremely high coefficients in the deviation choice variables for high coefficients in the auxiliary variables. However, since the auxiliary variables are unique to the conflict in question, their coefficients do not accumulate like those of the deviation variables. We reformulate all equality constraints in this way, but present the prior versions in Table 3 for notational brevity.

4 Numerical Experiments

In this section, we demonstrate the viability of the procedure, by performing deconfliction for a representative day of flights across the Atlantic.

4.1 Balancing Computational Feasibility and Quality of Solutions

4.2 North Atlantic Traffic Instance

4.3 Conservative Case Results

4.4 Lazy Constraint Generation Approach Demo

5 Discussion

5.1 Extensions

5.2 Conclusion

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Table 3: Temporal conflict case classification for all perturbable-perturbable (PP) and perturbable-nominal (PN) pairings

Conflict type	PP/PN	Constraint	Big-M infimum	Slack range(s)	Added bits
Safe	PP or PN	None	None	None	None
Temporally impossible	PP	$\mathbb{1}_{\kappa_1}^\top e^{(\kappa_1^{(1)})} + \mathbb{1}_{\kappa_2}^\top e^{(\kappa_2^{(1)})} = 0$	None	None	None
Temporally impossible	PN	$\mathbb{1}_{\sigma_1}^\top e^{(\sigma_1^{(1)})} = 0$	None	None	None
Force X	PP	$t^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)}, \widetilde{\kappa}_1^{(4)}) + \Delta - t^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)} - 1, \widetilde{\kappa}_2^{(4)})$ $-\mathcal{M}_X^{(\kappa)}(2 - \mathbb{1}_{\kappa_1}^\top e^{(\kappa_1^{(1)})} - \mathbb{1}_{\kappa_2}^\top e^{(\kappa_2^{(1)})}) + (B_-^{(\kappa)})^\top s_-^{(\kappa)} = 0$	$\mathcal{M}_X^{(\kappa)} = \beta_X^{(\kappa)}$	$[0, 2\mathcal{M}_X^{(\kappa)} - \alpha_X^{(\kappa)}]$	$\{s_-^{(\kappa)}\}$
Force X	PN	$t^{(\sigma_1^{(1)})}(\sigma_1^{(2)}, \sigma_1^{(3)}, \widetilde{\sigma}_1^{(4)}) + \Delta - t^{(\sigma_2^{(1)})}(\sigma_2^{(2)}, \sigma_2^{(3)} - 1, \cdot)$ $-\mathcal{M}_X^{(\sigma)}(1 - \mathbb{1}_{\sigma_1}^\top e^{(\sigma_1^{(1)})}) + (B_-^{(\sigma)})^\top s_-^{(\sigma)} = 0,$	$\mathcal{M}_X^{(\sigma)} = \beta_X^{(\sigma)}$	$[0, \mathcal{M}_X^{(\sigma)} - \alpha_X^{(\sigma)}]$	$\{s_-^{(\sigma)}\}$
Force Y	PP	$t^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)}, \widetilde{\kappa}_2^{(4)}) + \Delta - t^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)} - 1, \widetilde{\kappa}_1^{(4)})$ $-\mathcal{M}_Y^{(\kappa)}(2 - \mathbb{1}_{\kappa_1}^\top e^{(\kappa_1^{(1)})} - \mathbb{1}_{\kappa_2}^\top e^{(\kappa_2^{(1)})}) + (B_+^{(\kappa)})^\top s_+^{(\kappa)} = 0$	$\mathcal{M}_Y^{(\kappa)} = \beta_Y^{(\kappa)}$	$[0, 2\mathcal{M}_Y^{(\kappa)} - \alpha_Y^{(\kappa)}]$	$\{s_+^{(\kappa)}\}$
Force Y	PN	$t^{(\sigma_2^{(1)})}(\sigma_2^{(2)}, \sigma_2^{(3)}, \cdot) + \Delta - t^{(\sigma_1^{(1)})}(\sigma_1^{(2)}, \sigma_1^{(3)} - 1, \widetilde{\sigma}_1^{(4)})$ $-\mathcal{M}_Y^{(\sigma)}(1 - \mathbb{1}_{\sigma_1}^\top e^{(\sigma_1^{(1)})}) + (B_+^{(\sigma)})^\top s_+^{(\sigma)} = 0$	$\mathcal{M}_Y^{(\sigma)} = \beta_Y^{(\sigma)}$	$[0, \mathcal{M}_Y^{(\sigma)} - \alpha_Y^{(\sigma)}]$	$\{s_+^{(\sigma)}\}$
Choice	PP	$t^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)}, \widetilde{\kappa}_1^{(4)}) + \Delta - t^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)} - 1, \widetilde{\kappa}_2^{(4)})$ $-\mathcal{M}_X^{(\kappa)} z_\kappa - \mathcal{M}_X^{(\kappa)}(2 - \mathbb{1}_{\kappa_1}^\top e^{(\kappa_1^{(1)})} - \mathbb{1}_{\kappa_2}^\top e^{(\kappa_2^{(1)})}) + (B_-^{(\kappa)})^\top s_-^{(\kappa)} = 0,$ $t^{(\kappa_2^{(1)})}(\kappa_2^{(2)}, \kappa_2^{(3)}, \widetilde{\kappa}_2^{(4)}) + \Delta - t^{(\kappa_1^{(1)})}(\kappa_1^{(2)}, \kappa_1^{(3)} - 1, \widetilde{\kappa}_1^{(4)})$ $-\mathcal{M}_Y^{(\kappa)}(1 - z_\kappa) - \mathcal{M}_Y^{(\kappa)}(2 - \mathbb{1}_{\kappa_1}^\top e^{(\kappa_1^{(1)})} - \mathbb{1}_{\kappa_2}^\top e^{(\kappa_2^{(1)})}) + (B_+^{(\kappa)})^\top s_+^{(\kappa)} = 0$	$\mathcal{M}_X^{(\kappa)} = \beta_X^{(\kappa)},$ $\mathcal{M}_Y^{(\kappa)} = \beta_Y^{(\kappa)}$	$[0, 3\mathcal{M}_X^{(\kappa)} - \alpha_X^{(\kappa)}]$ $[0, 3\mathcal{M}_Y^{(\kappa)} - \alpha_Y^{(\kappa)}]$	$\{z_\kappa, s_-^{(\kappa)}, s_+^{(\kappa)}\}$
Choice	PN	$t^{(\sigma_1^{(1)})}(\sigma_1^{(2)}, \sigma_1^{(3)}, \widetilde{\sigma}_1^{(4)}) + \Delta - t^{(\sigma_2^{(1)})}(\sigma_2^{(2)}, \sigma_2^{(3)} - 1, \cdot)$ $-\mathcal{M}_X^{(\sigma)} z_\sigma - \mathcal{M}_X^{(\sigma)}(1 - \mathbb{1}_{\sigma_1}^\top e^{(\sigma_1^{(1)})}) + (B_-^{(\sigma)})^\top s_-^{(\sigma)} = 0,$ $t^{(\sigma_2^{(1)})}(\sigma_2^{(2)}, \sigma_2^{(3)}, \cdot) + \Delta - t^{(\sigma_1^{(1)})}(\sigma_1^{(2)}, \sigma_1^{(3)} - 1, \widetilde{\sigma}_1^{(4)})$ $-\mathcal{M}_Y^{(\sigma)}(1 - z_\sigma) - \mathcal{M}_Y^{(\sigma)}(1 - \mathbb{1}_{\sigma_1}^\top e^{(\sigma_1^{(1)})}) + (B_+^{(\sigma)})^\top s_+^{(\sigma)} = 0$	$\mathcal{M}_X^{(\sigma)} = \beta_X^{(\sigma)},$ $\mathcal{M}_Y^{(\sigma)} = \beta_Y^{(\sigma)}$	$[0, 2\mathcal{M}_X^{(\sigma)} - \alpha_X^{(\sigma)}]$ $[0, 2\mathcal{M}_Y^{(\sigma)} - \alpha_Y^{(\sigma)}]$	$\{z_\sigma, s_-^{(\sigma)}, s_+^{(\sigma)}\}$